ECE 310 – Wireless Communications · Milestone 3 Report

Quantifying the Sum-Rate Cost of Reciprocity in Beyond-Diagonal RIS-Aided MU-MISO Systems

A Manifold-Optimisation Comparison Across Single-, Group- and Fully-Connected Architectures

Abstract

A lossless passive BD-RIS has a unitary scattering matrix ($\Theta^H\Theta = I$). If it is also built only from reciprocal components, Θ must additionally be symmetric ($\Theta = \Theta^T$). This report quantifies the sum-rate lost by that extra constraint in a downlink MU-MISO system with a blocked direct link. We compare a reciprocal design (Stiefel-manifold conjugate-gradient ascent with a symmetry penalty and a final symmetric-unitary projection) against a general, non-reciprocal-capable baseline (the same optimiser with the penalty removed, plus the paper-faithful joint design of Li–Shen–Clerckx). Both receive identical channels, starting matrices, initial MMSE precoder and metric functions, so any difference is caused by the constraint.

Over 50 matched Rayleigh draws (N = K = 2, R = 32, Pmax = 20 dBm) the mean cost of reciprocity is exactly 0 for single-connected (SC), 0.103 bps/Hz (1.0 %, 95 % CI 0.057–0.159) for group-connected (GC, block size 4) and 0.021 bps/Hz (0.2 %, 95 % CI 0.001–0.046) for fully-connected (FC). The reciprocal design uses only 52–63 % of the real degrees of freedom of the general design and retains 99.0 % (GC) and 99.8 % (FC) of its rate. The gap is small but positive, so reciprocity is nearly free at this operating point, though not free. All results come from the M2 work; sweeps over RIS size, power, group size and fading are listed as open work.

1. Objectives of M3

M2 built the common simulation environment and one matched comparison. M3 must show effectiveness, compare against baselines and revise the SOTA position. Mapping to the M3 outcomes:

M3 outcome	Where addressed
1. Final method / implementation	Section 2
2. Effectiveness through experiments	Sections 3–4
3. Baseline comparison with a defined protocol	Sections 3.2, 4.1–4.5
4. Critical analysis (successes, failures, limits)	Section 5
5. Final contribution and revised SOTA positioning	Section 6

2. Final Method and Implementation

2.1 System model

BS with N antennas, BD-RIS with R ports, K single-antenna users, direct link blocked. Received signal $r = H_{RX}\,\Theta\,H_{TX}\,V\,s + n$, with $H_{TX} \in \mathbb{C}^{R\times N}$, $H_{RX} \in \mathbb{C}^{K\times R}$, $V \in \mathbb{C}^{N\times K}$, $\|V\|^2_F \le P_{max}$ and noise power N_o per user. With e_k the k-th row of $E = H_{RX}\,\Theta\,H_{TX}$ and $c_{ki} = e_k v_i$:

$\gamma_k = |c_{kk}|^2 / (\Sigma_{i\neq k} |c_{ki}|^2 + N_o)$, $R_{sum} = \Sigma_k \log_2(1 + \gamma_k)$ [bps/Hz]

Path loss $PL(d) = C_o(d/d_o)^{-\rho}$ with $C_o = -30$ dB, $\rho = 2.2$, BS–RIS 50 m, RIS–user 2.5 m, noise −80 dBm. Rayleigh fading, or Rician with K-factor 2 dB.

2.2 Architectures and feasible sets

Architecture	Θ structure	General feasible set (real DoF)	Reciprocal feasible set (real DoF)
SC	diagonal	R	R (diagonal is already symmetric)
GC	G blocks of size $R_g = R/G$	$G\cdot R_g^2$	$G\cdot R_g(R_g+1)/2$
FC	one R×R block	R^2	$R(R+1)/2$

SC is an exact-zero control: any non-zero SC gap would indicate an implementation error.

2.3 Reciprocal design

1. Common start: the shared diagonal SC matrix and MMSE precoder V_o.
2. Fractional programming (Lagrangian dual + quadratic transform) turns the sum rate into a surrogate with auxiliary variables τ (SINRs) and γ, refreshed after every accepted step.
3. Objective $F_{FP}(\Theta) - \nu\|\Theta - \Theta^T\|^2_F$ with $\nu = 1$; the analytic gradient includes the penalty term $-4\nu(\Theta - \Theta^T)$ and was checked against central finite differences (relative error $\approx 4.5 \times 10^{-9}$ for $R_g \in \{1, 4, 32\}$).
4. Riemannian gradient by tangent projection onto the unitary manifold; Polak–Ribière conjugate direction with reset; Armijo backtracking (contraction 0.75, $c = 2 \times 10^{-11}$, ≤ 200 steps); QR retraction with a positive-diagonal phase fix.
5. Stop when the surrogate changes by less than 10^{-8} or after 8000 iterations.
6. Exact enforcement: symmetrise each block, $\Theta_{sym} = \tfrac{1}{2}(\Theta_g + \Theta_g^T)$, then take the unitary polar factor via SVD. The polar factor of a symmetric matrix is symmetric, so the result is exactly unitary and symmetric (errors $\approx 10^{-14}$). Reported rates always use the projected matrix. The projection costs only about 1.1×10^{-3} bps/Hz on average.

2.4 General baseline

Two modes, both following Li, Shen and Clerckx (reflective mode, Algorithms 1 and 2):

- **Matched / V-frozen (`general_fixedV_optimizer`)** – the reciprocal CGA machinery with $\nu = 0$ and a plain block-unitary projection. This is the like-for-like baseline: the only difference from the reciprocal design is the symmetry constraint.
- **Joint (`li_joint_optimizer`)** – Li's block-coordinate ascent over (ι, τ, W, Θ) with closed-form/bisection precoder updates and Riemannian CG for Θ (≤ 300 outer, ≤ 30 inner iterations). It also updates V, so it is a reference ceiling and is **not** comparable with reciprocal fixed-V results.

2.5 Common integration

One configuration file, one channel generator (`rng(seed_base + d)`, fixed call order: channels → start matrix → precoder), one SINR/sum-rate implementation and one runner. Every reported rate is recomputed with the common metric functions. `tests/test_common_framework.m` verifies identical channels, start matrices, precoders and rates against the reciprocal code, the power constraint, reproducibility and configuration overrides.

3. Experimental Setup

3.1 Conditions

Parameter	Value
Antennas / users	N = K = 2
RIS ports	R = 32
Architectures	SC ($R_g = 1$), GC (G = 8 groups, $R_g = 4$), FC ($R_g = 32$)
Transmit power / noise	Pmax = 20 dBm, $N_o = -80$ dBm
Channel	Rayleigh; $C_o = -30$ dB, $\rho = 2.2$, BS–RIS 50 m, RIS–user 2.5 m
Optimiser	$\nu = 1$, tolerance 10^{-8}, 8000-iteration cap
Draws	50 matched draws, seeds 1–50 (MATLAB, committed in `M2/Results/matched_rayleigh_R32_50draws`)

All results use this single M2 configuration with the frozen-precoder protocol. Statistics not printed in the M2 README (medians, win counts, bootstrap CIs, mean iterations and run times) are computed from `per_draw.csv` of that run.

3.2 Metrics and protocol

- **Reciprocity gap** = general(fixed V) − reciprocal(fixed V), per draw (positive means reciprocity costs rate). Reported as mean ± standard error, a 95 % bootstrap CI of the paired mean (4000 resamples), the median, and the count of draws in which each design wins. Percent gap = mean gap / mean general rate.
- **Recomputed-V gap**: the same comparison after recomputing the MMSE precoder at the final Θ.
- **Feasibility**: $\|\Theta - \Theta^T\|_F$ and $\|\Theta^H\Theta - I\|_F$ of final matrices.
- **Effort**: optimiser iterations, cap hits, wall-clock time.

- **Baselines**: (i) general fixed-V (main), (ii) Li joint design (reference), (iii) SC control, (iv) common starting point.
- Negative gaps are reported, not clipped; both optimisers converge to local optima.

3.3 Provenance of the results

No new simulations were run for M3. Every number comes from the committed M2 results (the 50-draw run, plus the 3-draw quick check used only to confirm that seeds reproduce) and the M2 code documentation. Experiments beyond this operating point are listed in Section 8.

4. Results

4.1 Anchor result: 50 matched Rayleigh draws (condition A)

Mean ± SE, bps/Hz, frozen precoder:

Arch.	Reciprocal	General	Gap ± SE	95 % CI	Median	Gap %	General better / reciprocal better
SC	8.517	8.517	0.000	–	0	0	0 / 0
GC (R_g = 4)	9.909	10.012	0.103 ± 0.026	0.057–0.159	0.014	1.03	48 / 2
FC (R_g = 32)	10.310	10.331	0.021 ± 0.012	0.001–0.046	0.000	0.20	36 / 6 (8 ties)

With the precoder recomputed at the final Θ the gaps are 0.126 (GC) and 0.021 (FC) bps/Hz. Note that M2 quoted 0.88 % and 0.17 %, the average of per-draw percentages; the table uses mean gap divided by mean rate. Both are small.

4.2 Convergence effort and recomputed-V gap

Arch.	Reciprocal iters (mean)	General iters (mean)	Reciprocal time (s)	General time (s)	Reciprocal cap hits	Recomputed-V gap (bps/Hz)	General $\|\|\Theta - \Theta^T\|\|_F$ (mean)
SC	296	296	0.85	0.86	0 / 50	0.000	0
GC	1421	746	4.1	2.4	4 / 50	0.126	5.96
FC	570	649	2.6	3.6	0 / 50	0.021	2.76

Reciprocal GC needs about 1.9× the iterations of the general baseline and hit the 8000-iteration cap in 4 of 50 draws (21, 26, 31, 34); FC never did. Recomputing the MMSE precoder at the final Θ leaves the picture unchanged (GC 0.126, FC 0.021 bps/Hz).

4.3 Architecture comparison and complexity

Rate retained = reciprocal / general (frozen V). Real DoF from Section 2.2 (a hardware-count proxy, not a measured circuit cost):

Arch.	General DoF	Reciprocal DoF	DoF used	Rate retained
SC	32	32	100 %	100 %
GC (R_g = 4)	128	80	62.5 %	99.0 %
FC	1024	528	51.6 %	99.8 %

The fixed-V ordering SC < GC < FC holds for both designs. The reciprocal GC optimiser is the more expensive one (Section 4.2).

4.4 Joint-design reference

Li's joint design reaches 11.29 / 13.10 / 13.71 bps/Hz (SC / GC / FC, condition A), about 2.8–3.4 bps/Hz above the fixed-V baselines. That improvement comes from updating V, not from reciprocity, so it is a ceiling for later work, not a reciprocity gap. The joint design hit its 300-outer-iteration cap in nearly every draw, so these values are slightly conservative.

4.5 Validation of the implementation

- Reciprocal solutions are symmetric and unitary to about 10^{-14} after the final projection; general solutions are unitary to about 10^{-14} or better.
- General GC and FC solutions are genuinely non-symmetric (mean $\|\Theta - \Theta^T\|_F \approx 6.0$ for GC, 2.8 for FC), so they use the freedom reciprocity removes.
- SC gives identical results from both pipelines in all 50 draws (exact-zero control passed).
- Matched seeds reproduce draws 1–3 of the quick check inside the 50-draw run exactly.
- The analytic gradient matches central finite differences to a relative error of about 4.5×10^{-9}.

5. Critical Analysis

What worked. (1) The controlled protocol isolates the constraint: the SC control is exactly zero and feasibility holds at machine precision. (2) The GC cost is positive in 48 of 50 draws and its confidence interval excludes zero. (3) The FC cost is tiny: median zero, mean 0.2 % of the rate.

Unexpected result: FC gap < GC gap. FC removes far more degrees of freedom than GC (48 % vs 37.5 % fewer), yet loses less rate. A plausible but **untested** explanation is that with N = K = 2 the system only needs a low-rank effective channel; a 32×32 symmetric-unitary matrix has ample freedom to realise it, whereas GC blocks of size 4 (= N + K) are small enough that symmetry binds. This is consistent with the low-rank result reported for symmetric-unitary RIS (Santamaria et al., arXiv 2604.22469, read by us only via an abstract-level summary). A group-size sweep (R_g = 8, 16) and an N = K sweep would test it.

Where the numbers should be read with care.

- *Optimiser quality vs constraint.* Both methods stop at local optima, and the reciprocal GC optimiser hit its iteration cap in 4 of 50 draws. The GC gap therefore partly reflects convergence

under the penalty formulation, and an exact symmetric-unitary method would be needed to separate the two effects.

- *Negative gaps.* Reciprocal beat general in 2 of 50 GC and 6 of 50 FC draws (up to 0.27 bps/Hz). This is local-optimum variability, because the global general optimum can never be below the reciprocal one.
- *Heavy-tailed FC gap.* The mean (0.021) is driven by a few draws (37, 41, 46 with 0.27–0.38 bps/Hz); the median is zero. Use the median and CI, not only the mean.
- *Single operating point.* Only R = 32, 20 dBm, Rayleigh and N = K = 2 were measured. How the gap depends on RIS size, power, fading and group size is not known from our data and should not be inferred.

Known failure kept visible. The reciprocal FC mean (10.31 bps/Hz, 10.87 with V recomputed) is below the ≈ 14 bps/Hz of the source reciprocal paper's FC curve. We did not tune parameters to close this. The likely causes are the frozen precoder (the joint design reaches 13.7) and an unstated power convention, neither of which has been verified.

6. Baseline Comparison and Revised SOTA Positioning

6.1 Against the closest prior work

	Li et al. 2023	Fidanovski 2025 (2509.20246)	Fidanovski 2607.22170	This project
Reciprocity enforced	no	yes (penalty)	yes (exact)	both pipelines side by side
Non-reciprocal benchmark in same study	n/a	no	yes (graphical)	yes, numeric
Beamformer	joint	MMSE / uniform	isotropic	identical fixed MMSE + joint reference
Reported output	rates	rates	curves	gap in bps/Hz and %, CI, win counts
Fading	Rayleigh + Rician	both	not stated	Rayleigh (Rician not yet run)
Sweeps	P, M	P, R	R only	one point (R = 32, 20 dBm); sweeps open

6.2 Status of the hypotheses stated in the SOTA notes

Hypothesis	Status after M3
H1: quantified gap (bps/Hz, %, CI) across M, G,	**Partly answered.** Quantified with CIs at one point (R

Hypothesis	Status after M3
Pmax	= 32, 20 dBm, GC block size 4 and FC). No sweep over M, G or Pmax.
H2: gap under a fixed MMSE beamformer	**Answered** for N = K = 2 (recomputed-V gap also reported).
H3: Rayleigh and Rician; N = K up to 6; M ≤ 64	**Not answered.** Rayleigh, N = K = 2, R = 32 only.
H4: sensitivity to how reciprocity is enforced	**Not answered.** Only penalty + projection was run; the iteration-cap hits show why it matters.
H5: performance–complexity trade-off	**Partly answered.** DoF-based trade-off in Section 4.3; no circuit-count or runtime Pareto.

6.3 Revised positioning

The M1 claims that the question is unanswered or that this is a first characterisation are withdrawn; arXiv 2607.22170 already reports near-coincident curves. Supported by our evidence, the contribution is narrower:

> *Under a common fixed MMSE beamformer and matched Rayleigh channels (N = K = 2, R = 32, 20 dBm), imposing $\Theta = \Theta^T$ on a lossless BD-RIS costs exactly 0 for SC, about 0.10 bps/Hz (≈ 1 % of the rate) for GC with block size 4 and about 0.02 bps/Hz (≈ 0.2 %) for FC, while using 52–63 % of the degrees of freedom of the general design. The cost is statistically distinguishable from zero for GC and borderline for FC.*

This agrees in direction with the earlier graphical result, adds numbers and uncertainty, and adds the unexpected GC > FC ordering together with the optimiser-convergence caveat.

7. Final Contribution

1. A reproducible matched framework (common config, channels, metrics, runner, tests) in which reciprocal and general designs differ only in the symmetry constraint.
2. Numerical evidence of the reciprocity gap with confidence intervals for SC, GC and FC at the M2 operating point.
3. A degrees-of-freedom vs rate-retained comparison for SC, GC and FC.
4. Documented limits: iteration-cap effects, local-optimum negative gaps, the unresolved FC reproduction gap, and what the joint-design ceiling means.

8. Limitations and Future Work

- **Single operating point.** Only R = 32, Pmax = 20 dBm, Rayleigh fading and N = K = 2 were evaluated (50 draws). Sweeps over R, Pmax, G, N = K and Rician fading were planned for M3 but are not part of this report, so no claim is made beyond this point.
- Penalty + projection is the only enforcement method; an exact symmetric-unitary manifold (Takagi retraction) would separate constraint cost from optimiser error.

- Frozen precoder; a joint reciprocal design (V and Θ) is not implemented.
- Complexity is a DoF proxy plus iteration counts; no circuit-level or asymptotic cost analysis.
- Li's Algorithm 3 (closed-form SC) and hybrid/STAR modes are out of scope.

9. Individual Contributions

Member	Contribution
Samriddhi	*literature, SOTA synthesis, mathematical formulation, report integration*
Aashi	Reciprocal pipeline: FP surrogate, analytic gradient and finite-difference check, tangent projection, retraction, CGA + Armijo, symmetric-unitary projection, convergence and Monte-Carlo scripts
Kahaan	General non-reciprocal baseline: Li Algorithms 1 and 2 (`li_*` functions), V-frozen matched optimiser, block-unitary projection, general-baseline tests and trend validation
Prakshal	Common configuration, channel and metric functions, matched runner, result storage, plots, integration tests, 50-draw run

10. Reproducibility

- M2 data: `M2/Results/matched_rayleigh_R32_50draws/` (`per_draw.csv`, `summary.csv`, `config_used.txt`, figures) and the 3-draw quick check.
- Code: `M2/Code/` (`reciprocal`, `nonreciprocal`, `Common_simulation_and_integration`) with the documentation in each folder's README.
- Settings: v = 1, tolerance 10^{-8}, 8000-iteration cap, Armijo c = 2×10^{-11} (reciprocal) and 10^{-4} (Li inner loop), seeds 1–50.
- The extra statistics in this report (medians, win counts, bootstrap CIs) are computed from `per_draw.csv`.

References

[1] H. Li, S. Shen, B. Clerckx, "Beyond Diagonal RIS: From Transmitting and Reflecting Modes to Single-, Group-, and Fully-Connected Architectures," IEEE TWC 22(4):2311–2324, Apr. 2023 (arXiv 2205.02866).

[2] M. Fidanovski et al., "Reciprocal BD-RIS: Scattering Matrix Design via Manifold Optimization," arXiv 2509.20246 (journal metadata in the M1 list is unverified).

[3] M. Fidanovski, Venkataramanaiah, Jorswieck, de Abreu, arXiv 2607.22170 (read at abstract level only).

[4] I. Santamaria et al., arXiv 2604.22469 (read at abstract level only).

[5] S. Shen, B. Clerckx, R. Murch, "Modeling and Architecture Design of RIS Using Scattering Parameter Network Analysis," IEEE TWC 21(2):1229–1243, 2022.

[6] K. Shen, W. Yu, "Fractional Programming for Communication Systems – Part I," IEEE TSP 66(10):2616–2630, 2018.

[7] N. Boumal, An Introduction to Optimization on Smooth Manifolds, Cambridge University Press, 2023.